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The system power (P) should not include any power associated with devices which are not used to fulfil the lighting function, even if they are connected to the same network.

F.2.4 Area (A_i) to be used for calculation of the power density indicator

The area used for calculation of the power density indicator (D_P) shall be identical to the area used in lighting design for lighting calculation of parameters according to Clause 4.1.1 and described in AS/NZS 1158.2.

F.2.5 Power density indicator (PDI)—Calculation worked examples

F.2.5.1 Worked example of a lighting scheme without constant light output (CLO)

A local area road designed to AS/NZS 1158.3.1 Category P, with LED luminaires with off-peak adaptive lighting via central management system (CMS) controls.

Peak demand operation state to be in accordance with subcategory PR3 (required maintained average horizontal illuminance 1.75 lx), and off-peak demand operation state to be in accordance with subcategory PR5 (required maintained average horizontal illuminance 0.85 lx) (e.g. midnight to 6 a.m., say 0.5 of operating hours).

Lighting scheme physical parameters (numerical values are indicative only):

(a) Road length = 1000 m.

(b) Road reserve width (RRW) = 18 m.

(c) Luminaire spacing = 50 m.

(d) Luminaire quantity = 20.

(e) Luminaire power (light source and light source control gear) = 20 W.

(f) Light point controller power (e.g. NEMA/ANSI C136.41 device) = 2 W, one per luminaire.

(g) System device power (e.g. RF controls communication gateway) = 30 W, one per scheme.

In accordance with Paragraph F2.1 above, where the lighting subcategory changes during the night, the power density (D_P) should be calculated separately for each of the lighting subcategories, and may be calculated as an average over this period.

The system power (P) calculation for the above example is—

$$P = \sum_{k=1}^{n_{lp}} P_k + P_{ad} \qquad \text{F3}$$

where

P = total system power of the lighting scheme (W)

P_k = operational power of light point = (20 + 2) = 22 W

P_{ad} = total operational power of system devices = 30 W

Therefore the system power $(P) = (20 \times 22) + (30) = 470\,W$.

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Printed / viewed by: [s225048421@deakin.edu.au] @ 2026-03